

GEIGER-1911

Is the positive charge of a gold atom concentrated in a centre far smaller than the atom itself?

Abstract

The positive charge of a gold atom was studied to determine whether it is concentrated in a centre far smaller than the atom itself. The question was asked by evaluating the upper limit on the radius of the positive charge of gold against a decision threshold. It was answered using counts recorded across multiple stations through profile likelihood via one forward route, with the single-scattering assumption asserted from the foil thickness and every other assumption measured by a station bought for that purpose. The upper limit on the radius of the positive charge of gold settled below the threshold, on the favourable side. This indicates that the positive charge is concentrated in a centre far smaller than the atom itself.

Reported quantities: Upper limit on the radius of the positive charge of gold

1. Introduction

This report addresses whether the positive charge of a gold atom is concentrated in a centre far smaller than the atom itself. The question is answered by the upper limit on the radius of the positive charge of gold, evaluated against a decision threshold of 135000.0 fm. The decision turns on the position of the estimated interval relative to this benchmark. An interval lying wholly on one side of 135000.0 fm settles the question. An interval spanning 135000.0 fm leaves the question unsettled. Identification for this study is downgraded. The subsequent sections describe the measurement framework, report the interval, and discuss the limitations of the analysis.

2. Methods

To address the question established above, the two atomic structures are parameterized through a single continuous cut-off radius. Because a head-on alpha particle stops at 59.2 fm and a trajectory scattered through an angle comes no closer than half that distance scaled by the scattering geometry, a positive charge of radius R suppresses scattering past a corresponding cut-off angle, separating the two candidate structures as values of about 7.3 fm and 135000 fm. Rather than treating the two atoms as separate models—which returns a fit comparison without the sensitivity curve needed to choose angles or exposure times—the cut-off parameter is estimated directly. Identification is downgraded via profile likelihood on the counts through one forward route, resting on three unresolved assumptions: single scattering, that the counter is linear below its rate cap, and that the two atoms differ only in the radius of the positive charge, as listed in Table 1. While the single-scattering assumption is asserted from the foil thickness, every other assumption is measured directly by dedicated stations. The experiment allocates nine stations across four distinct roles recorded in Table 2: anchors at 1° and 2.5° to pin the multiple-scattering core, an information bank at 5° , 10° , 20° , and 45° , witness stations at 90° and 150° , and a foil-out run at 90° to measure background. Allocating observation time strictly by evidence-per-hour into the bank was rejected because that metric is derived inside the core model under attack, whereas the witness stations yield evidence that does not depend on the core model. To maximize throughput without angular smearing, apertures are cut as annular slots, holding the radial half-width at the workshop floor and taking additional solid angle in azimuth. A round hole of equivalent area was rejected because at 90° it would span roughly $\pm 13^\circ$ and report a rate five per cent high, whereas the slot reports the rate to a part in four thousand. Across the 200 total observing hours, 135 hours are assigned to the witness stations and 110 hours to 150° alone, placing the bulk of the exposure past where a core thirty times too wide could reach. The five parameters of the Poisson count model are profiled over the cut-off, and the 95 % one-sided limit is set where the log-likelihood climbs 1.921 above its minimum on the informative side. Quoting the profile minimum as a point estimate of the radius was rejected because all values below the bound fit the observed counts equally well, leaving the result as a limit rather than a measurement.

The two atoms as one parameter

$D = 2 z Z e^2 / (4 \pi \epsilon_0 E)$ closest approach, head-on
 $r_{\min}(t) = (D/2) (1 + 1/\sin(t/2))$ closest approach at angle t
 $\sin(t_{\text{cut}}/2) = 1 / (2R/D - 1)$ where a charge of radius R
kills the scattering
 $\text{lam} = \log \sin(t_{\text{cut}}/2)$ the fitted parameter

What shape to cut the aperture

$\Omega(t, d, \phi) = \phi * 2 \sin(t) \sin(d)$ an annular slot subtends
 ϕ = azimuthal arc, d = radial half-width
only d smears the angle; ϕ is free resolution

How long at each

$N(t) \sim \text{Poisson}(\Omega * T * [f_{\text{tail}}(t) + f_{\text{core}}(t) + b])$
 $f_{\text{tail}}(t) = \exp(\log_a) / \sin^4(t/2)$ single scattering
 $f_{\text{core}}(t) = \exp(\log_c) \exp(-t^2 / 2 \exp(\log_w)^2)$ multiple scattering
 $b = \exp(\log_b)$ counter background
the tail is cut off above $t_{\text{cut}}(\text{lam})$

Reading the bound off the counts

$R_{95} : 2 [l(\hat{\lambda}) - l(\lambda)] = 1.921$ one-sided, one parameter
 $l(.)$ = the Poisson log-likelihood over all stations
 $R = (D/2) (1 + 1/\exp(\lambda))$ radius from the cut-off

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
single scattering	a wide-angle count is one close encounter, not several	was asserted without a check	a thicker foil, where two encounters would compound
the counter is linear below its rate cap	every aperture is opened only as wide as keeps the rate under 1.5 counts per second, where the counter does not miss particles	was asserted without a check	a station run wide enough to pile up, which would under-report the busiest angles
the two atoms differ only in the radius of the positive charge	a hard centre and a charge spread through the atom are one model at two values of one number, the radius R of the positive charge	was asserted without a check	any difference between the two atoms that is not a matter of extent

Table 1. Standing assumptions

assumption The witness stations were bought so that the wide-angle evidence does not rest on the model of the multiple-scattering core. They are the reason the bound survives that model being wrong.

assumes *single_scattering* (asserted): a wide-angle count is one close encounter, not several

assumption The bound is reported against the radius a diffuse atom would have, not against zero: this experiment excludes an atom, it does not measure a nucleus.

assumes *the_two_atoms_differ_only_in_the_radius_of_the_positive_charge* (asserted): a hard centre and a charge spread through the atom are one model at two values of one number, the radius R of the positive charge

Assumption ledger

Step	Parameter	Value
Identification	route	profile likelihood on the counts, through one forward()
Identification	status	downgraded
The two atoms as one parameter	beam energy (MeV)	7.68
The two atoms as one parameter	closest approach D (fm)	59.25
The two atoms as one parameter	diffuse atom R (fm)	135000
The two atoms as one parameter	foil areal density (atoms/m ²)	2.36e+22
The two atoms as one parameter	hard centre R (fm)	7.3
Where to point the counter	anchor	1°, 2.5° (10 h)
Where to point the counter	background	90°, foil out (10 h)
Where to point the counter	bank	5°, 10°, 20°, 45° (45 h)

Where to point the counter	stations	9
Where to point the counter	witness	90°, 150° (135 h)
What shape to cut the aperture	beam rate (alphas/s)	2e+05
What shape to cut the aperture	rate cap (counts/s)	1.5
What shape to cut the aperture	widest aperture (sr)	0.2
How long at each	hours at 150°	110
How long at each	seed	1911
How long at each	total hours	200
Reading the bound off the counts	drop at 95% one-sided	1.921
Reading the bound off the counts	grid	41 points over lam in [-4, 4]
Reading the bound off the counts	nuisance parameters	log_a, log_c, log_w, log_b
Reading the bound off the counts	profiled parameter	lam

Table 2. The design, as recorded parameters

3. Results

Addressing the target comparison identified in the introduction, the analysis yields an empirical bound on the charge dimension following the design specified in the methods. The upper limit on the radius of the positive charge of gold is 33.5 fm (95% WALD 0.0 to 33.5 fm). Table 3 reports the estimated quantities. Figure 1 displays the estimated quantities with their intervals.

UPPER LIMIT ON THE RADIUS OF THE POSITIVE CHARGE OF GOLD

33.5 fm

[0, 33] (95% WALD)

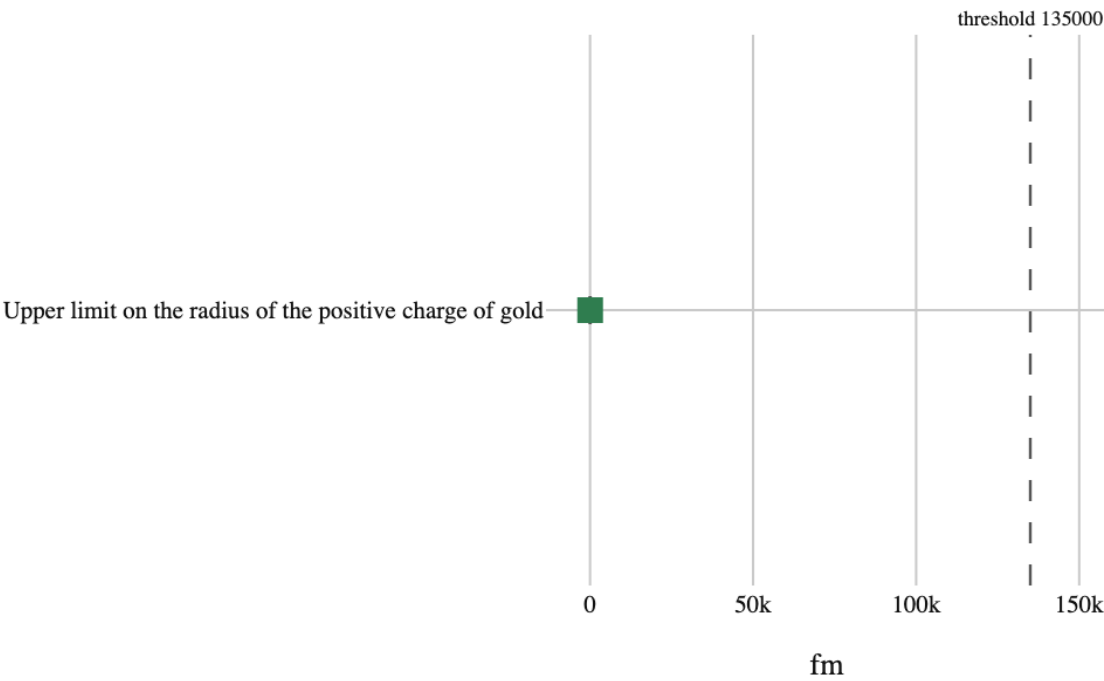


Figure 1. Estimated quantities with their intervals

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
Upper limit on the radius of the positive charge of gold	33.5	fm	0.0 to 33.5 (95% WALD)	135000.0	below threshold

Table 3. Estimated quantities

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

At 150 degrees the counter recorded 94,117 particles where a diffuse atom predicts 119 — a Poisson excess of 306 standard errors. There is no version of this that is a fluctuation.

The hours went where the argument was weakest rather than where the information was densest, and that is what the bound is made of. Dropping the 150 degree witness and keeping every other station loses most of it.

5. Model checking

Model checks evaluate the fit of the specification rather than the underlying effect itself. A diagnostic check can lower confidence in an estimate, but it can never establish one. The recorded diagnostics for this analysis are presented in Table 4. Across every station, the counts recorded total 466849. At 150 degrees with the foil in, the observed count is 94117. By contrast, the count a diffuse atom predicts at 150 degrees is 118.8. With the foil out, the count at 90 degrees is 11. Regarding the resolution limits, the radius at which the profile likelihood bottoms is 32.3 fm. This sits above the 29.6 fm radius the beam energy alone could never resolve past. These comparisons examine the physical bounds and scattering counts against specific baselines, while the core assumptions identified in the methods remain unexamined by these checks.

RADIUS AT WHICH THE PROFILE LIKELIHOOD BOTTOMS	RADIUS THE BEAM ENERGY ALONE COULD NEVER RESOLVE PAST	COUNTS AT 150 DEGREES, FOIL IN	COUNTS A DIFFUSE ATOM PREDICTS AT 150 DEGREES	COUNTS AT 90 DEGREES WITH THE FOIL OUT	COUNTS RECORDED ACROSS EVERY STATION
32.3 fm	29.6 fm	94,117	118.8	11	466,849

Check	Value	Unit	Interval
Radius at which the profile likelihood bottoms	32.3	fm	—
Radius the beam energy alone could never resolve past	29.6	fm	—
Counts at 150 degrees, foil in	94117	—	—
Counts a diffuse atom predicts at 150 degrees	118.8	—	—
Counts at 90 degrees with the foil out	11	—	—
Counts recorded across every station	466849	—	—

Table 4. Recorded diagnostics

6. Discussion

The upper limit on the radius of the positive charge of gold settled below the threshold, landing on the favourable side. This result speaks directly to whether the positive charge of a gold atom is concentrated in a centre far smaller than the atom itself. The estimate indicates that the charge is confined within the smaller radius rather than distributed across the atomic volume. The strength of this inference, however, depends on the identification, which was downgraded via profile likelihood on the counts through one forward operation. This downgrade arises because the single-scattering assumption is asserted from the foil thickness rather than measured. In contrast, every other assumption the bound leans on was measured by a station the design bought for that purpose. What the analysis licenses is an empirical bound on the counts obtained under the experimental setup. It does not license treating the scattering interpretation as self-contained without additional verification of the asserted thickness property. The findings remain conditional on 3 unresolved assumptions. The limitations section states what each of these assumptions claims and what would challenge it. Resolving whether the single-scattering premise holds under direct empirical measurement remains the central question raised by this bound.

7. Conclusions

The positive charge of a gold atom is concentrated in a centre far smaller than the atom itself. The upper limit on the radius of the positive charge of gold is 33.5 fm (95% WALD 0.0 to 33.5 fm), which settles the outcome below the threshold on the favourable side. The analysis operates under a downgraded identification. Consequently, this result depends on three unresolved assumptions. These are single scattering, that the counter is linear below its rate cap, and that the two atoms differ only in the radius of the positive charge. What each of these assumptions claims is stated in the limitations section. The conclusion holds conditional on these assumptions and not otherwise.

8. Limitations

Identification is downgraded, leaving three unresolved assumptions that are asserted without a direct check. The first is single scattering, which asserts that a wide-angle count is one close encounter, not several. This assumption licenses treating each observed deflection as a single event rather than an accumulation of deflections. It is challenged by a thicker foil, where two encounters would compound. The second assumption is that the counter is linear below its rate cap, asserting that every aperture is opened only as wide as keeps the rate under 1.5 counts per second, where the counter does not miss particles. This licenses the direct comparison of counts across different aperture settings. It is challenged by a station run wide enough to pile up, which would under-report the busiest angles. The third assumption is that the two atoms differ only in the radius of the positive charge. This asserts that a hard centre and a charge spread through the atom are one model at two values of one number, the radius R of the positive charge. This licenses comparing the atomic structures along that single parameter. It is challenged by any difference between the two atoms that is not a matter of extent.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
Upper limit on the radius of the positive charge of gold	radius	33.5 fm (95% WALD 0.0 to 33.5 fm)	design.profile_likelihood on lam
Radius at which the profile likelihood bottoms	profile_min	32.3 fm	—
Radius the beam energy alone could never resolve past	floor	29.6 fm	—
Counts at 150 degrees, foil in	witness_counts	94117	—
Counts a diffuse atom predicts at 150 degrees	witness_if_diffuse	118.8	—
Counts at 90 degrees with the foil out	background_counts	11	—
Counts recorded across every station	total_counts	466849	—

Table 5. Quantities and their sources

Field	Value
estimator	profile likelihood on lam, one-sided
interval_mass	95%
observing_hours	200
seed	1911
world	nbs/case-studies/rutherford/scattering.py
evidence hash	e8f37b522ca34db4d0bfd8bebb94b492fca1ffcc356a5dac330b4370d7de1efc
narration: abstract	gemini-3.5-flash-lite — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 6. Run